In-class exercise 6 (solution)

Let's consider a monoclinic crystal system

Real: {1, 1, 1, 90, 45, 90}

Compute the interplanar spacing for (102) planes.

$$\left|\vec{k}^*_{hkl}\right|^2 = \frac{1}{d^2_{hkl}} = \frac{h^2}{a^2\sin^2\beta} + \frac{k^2}{b^2} + \frac{l^2}{c^2\sin^2\beta} - \frac{2hl\cos\beta}{ac\sin^2\beta}$$

$$\left|\vec{k}^*_{102}\right|^2 = \frac{1}{d^2_{102}} = \frac{1^2}{1^2\sin^2 45} + \frac{0^2}{1^2} + \frac{2^2}{1^2\sin^2 45} - \frac{2\times 1\times 2\cos 45}{1\times 1\sin^2 45} = 10 - 4\sqrt{2}$$

Alternative method (metric tensor) - faster:

$$\Gamma^*_{monocl.} = \begin{bmatrix} \frac{1}{a^2\sin^2\beta} & 0 & \frac{-\cos\beta}{ac\sin^2\beta} \\ 0 & \frac{1}{b^2} & 0 \\ \frac{-\cos\beta}{ac\sin^2\beta} & 0 & \frac{1}{c^2\sin^2\beta} \end{bmatrix}$$

$$\left|\vec{k}^*_{102}\right|^2 = [1 \quad 0 \quad 2]\begin{bmatrix} 2 & 0 & -\sqrt{2} \\ 0 & 1 & 0 \\ -\sqrt{2} & 0 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = 10 - 4\sqrt{2}$$

$$d_{102} = {}^{1}\!\Big/\!\left|\vec{k}^*_{102}\right| = 0.488$$